Project: Worked examples
Title: Accumulator sizing (gas law)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The pre-charge gas volume V0 of a hydraulic accumulator that delivers, or absorbs, a liquid volume ΔV between the lowest working pressure p1 and the highest p2, from $p \cdot V^n$ = const with absolute pressures and an exponent n between 1 (isothermal) and 1.4 (adiabatic); the volume scaled to the lowest operating temperature (T_max/T_min) and the pre-charge pressure at the charging temperature; the next size from your own size list; and the bar·litre product PS·V of the vessel's own maximum allowable pressure and volume against the PSSR Schedule 1 250 bar·L exception. Three modes: energy storage, shock and pulsation (screening only) and emergency stroke (ΔV is the stroke volume, linkable from the cylinder circuit). Not a pressure-vessel code check.
Absolute pressures: p_a = p_gauge + p_atm (or as entered, in Absolute mode) Gas volume: V0 = ΔV / [(p0/p1)^(1/n) − (p0/p2)^(1/n)], n = 1 isothermal … 1.4 adiabatic, then × the real-gas factor f_rg (user input, at least 1) and × (T_max + 273.15)/(T_min + 273.15), because p0 is defined at T_max and the gas delivers less when cold Pre-charge rules (records, applied to absolute pressures): p0 ≤ r_0·p1 at T_max (default 0.9, bladder type) and p2/p0(T_min) ≤ r_p (default 4), p0(T_min) = p0·(T_min + 273.15)/(T_max + 273.15) Pre-charge at the charging temperature: p0(T_c) = p0(T_max)·(T_c + 273.15)/(T_max + 273.15) Next size: the smallest listed size not below V0 Bar·litres: PS·V, PS = the vessel's maximum allowable pressure (bar gauge, entered, not below p2 gauge), V = the vessel volume entered (default the selected size); compared with 250 bar·L (PSSR Schedule 1, indication only; not available without PS and a volume)
Ref: Pressure Systems Safety Regulations 2000 (SI 2000/128) Schedule 1 — 250 bar·litre exception, indication only
Ref: Pressure Equipment Directive 2014/68/EU and UK PER 2016 — Pressure equipment; not checked here
Ref: Esposito, Fluid Power with Applications Ch. on accumulators — Gas-charged accumulator sizing, $p \cdot V^n$ = const

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Duty	Energy storage		assumption: Slow clamp-and-hold cycle: stored-energy duty.	any
dV	Liquid volume ΔV delivered, absorbed or (emergency stroke) needed for the stroke	14	L	spec: Hydraulic specification HS-12: clamp stroke plus leakage make-up, 14 L per cycle	0.001 … 1000 L
p_basis	Pressures entered as	Absolute		assumption: Pressures entered as absolute bar, as the source numbers are quoted.	any
p_0	Gas pre-charge pressure p0, at the maximum operating temperature	90	bar	spec: Hydraulic specification HS-12: pre-charge 90 bar at the maximum oil temperature	0 … 700 bar

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_1	Lowest working pressure p1	100	bar	spec: Hydraulic specification HS-12: lowest clamp pressure 100 bar	0 … 700 bar
p_2	Highest working pressure p2	210	bar	spec: Hydraulic specification HS-12: pump cut-out 210 bar	0 … 700 bar
p_atm	Atmospheric pressure (gauge to absolute)	1.01325	bar	assumption: Standard atmosphere, derived constant.	0.5 … 1.1 bar
n	Gas exponent n (1 isothermal, slow; 1.4 adiabatic, fast)	1		assumption: Isothermal: the clamp holds for minutes between cycles.	1 … 1.4
f_rg	Real-gas factor on V0 (1 = ideal gas; at least 1, as nitrogen's compressibility factor exceeds 1 at high pressure; from a gas-property source above 200 bar)	1		assumption: Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.	1 … 1.5
T_max	Maximum operating temperature (p0 applies here)	50	degC	spec: Hydraulic specification HS-12: oil temperature limit 50 °C	-40 … 150 degC
T_min	Minimum operating temperature (the gas is coldest and delivers least here; V0 is scaled by T_max/T_min)	30	degC	assumption: Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.	-40 … 150 degC
T_c	Charging temperature	20	degC	assumption: Charged in the workshop at 20 °C.	-40 … 100 degC
r_0	Pre-charge rule: p0 (at T_max) at most this fraction of p1, on absolute pressures (state whether the manufacturer's ratio is on gauge or absolute pressures)	0.9		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	0.1 … 1
r_p	Pre-charge rule: p2/p0 (p0 at T_min) at most this ratio, on absolute pressures	4		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	1 … 20

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
sizes	Available accumulator sizes (from the manufacturer's range)	4 row(s)		reference: record: illustrative size list invented for this example, to be replaced by the manufacturer's range	any
sizes[1]		V = 10 L			
sizes[2]		V = 20 L			
sizes[3]		V = 32 L			
sizes[4]		V = 50 L			
PS	Vessel maximum allowable pressure PS, bar gauge (nameplate or relief setting; not below p2 gauge; needed for the PSSR indication)	230	bar	spec: Hydraulic specification HS-12: accumulator relief setting 230 bar gauge (vessel PS; illustrative, to be replaced by the nameplate)	0 ... 1500 bar

3. Assumptions

A1. All gas-law pressures are absolute. A gauge entry is converted with the atmospheric pressure entered here, so the same numbers read as gauge or as absolute give different volumes. ($p \cdot V^n$ = const on a fixed mass of gas.)
A2. The exponent n is the user's: 1 for a slow discharge with heat exchange (isothermal), towards 1.4 for a fast one (adiabatic). The larger n gives the larger volume for a given ΔV. (Gas-law exponents, closed form.)
A3. The gas is ideal unless a real-gas factor is entered (at least 1: nitrogen's compressibility factor exceeds 1 at high pressure, so the real gas needs more volume, never less); above 200 bar the sheet warns that nitrogen departs from the ideal-gas law and a factor from a gas-property source should be entered. (Real-gas correction as a user factor.)
A4. The pre-charge p0 is defined at the maximum operating temperature T_max and V0 is sized there; at the lowest operating temperature T_min the same gas delivers only the fraction T_min/T_max (absolute) of ΔV, so V0 is multiplied by T_max/T_min, and the p2/p0 rule is checked with p0 at T_min. The temperature range is the user's; with T_min = T_max the correction is 1. (Manufacturer sizing practice; the gas law at constant mass.)
A5. The pre-charge rules ($p0 \le 0.9 \cdot p1$ and $p2/p0 \le 4$ by default) are records for the accumulator type and make and are checked as PASS/FAIL on ABSOLUTE pressures, whichever basis the pressures are entered on; the right values, and whether the manufacturer states the ratio on gauge or absolute pressures, come from the manufacturer (the same numbers read as gauge pressures give a different ratio). (Rules are records, not built-in limits.)
A6. The bar·litre product is PS times V of the vessel itself: PS is the vessel's maximum allowable pressure in bar gauge (the nameplate or relief setting, never below the highest working pressure p2) and V its internal volume (the selected size unless entered). Without PS and a volume the indication is not available. It indicates whether the 250 bar·L exception of PSSR Schedule 1 can apply; it is not a compliance statement, and PED / UK PER, PSSR examination and the vessel's own design code stay with the manufacturer and the competent person. (Addendum B §B5 regulatory gating: no pressure-vessel code check.)
A7. mode (Duty) is taken as Energy storage. (Slow clamp-and-hold cycle: stored-energy duty.)

A8. p_basis (Pressures entered as) is taken as Absolute. (Pressures entered as absolute bar, as the source numbers are quoted.)
A9. p_atm (Atmospheric pressure (gauge to absolute)) is taken as 1.01325 bar. (Standard atmosphere, derived constant.)
A10. n (Gas exponent n (1 isothermal, slow; 1.4 adiabatic, fast)) is taken as 1. (Isothermal: the clamp holds for minutes between cycles.)
A11. f_rg (Real-gas factor on V0 (1 = ideal gas; at least 1, as nitrogen's compressibility factor exceeds 1 at high pressure; from a gas-property source above 200 bar)) is taken as 1. (Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.)
A12. T_min (Minimum operating temperature (the gas is coldest and delivers least here; V0 is scaled by T_max/T_min)) is taken as 30 degC. (Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.)
A13. T_c (Charging temperature) is taken as 20 degC. (Charged in the workshop at 20 °C.)
A14. Energy storage duty: ΔV is the liquid volume the accumulator delivers between p_2 and p_1 over a cycle; n = 1 for a slow discharge, towards 1.4 for a fast one. (Gas-law sizing for a stored-energy duty.)

4. Working

[15] pa_0 = p_0
= 90 bar
= 90 bar
Pre-charge pressure, absolute

[16] pa_1 = p_1
= 100 bar
= 100 bar
Lowest working pressure, absolute

[17] pa_2 = p_2
= 210 bar
= 210 bar
Highest working pressure, absolute

[18] p2_g = max(pa_2 - p_atm, 0 bar)
= max(210 bar - 1.0132 bar, 0 bar)
= 208.99 bar
Highest working pressure, gauge

[20] V0_ideal = dV / [(pa_0/pa_1)^(1/n) - (pa_0/pa_2)^(1/n)] $p \cdot V^n = \text{const}$
= 14 L / [(0.9) - (0.42857)]
= 29.697 L
Gas volume for the displacement, ideal gas

[21] f_T = (T_max / 1 K) / (T_min / 1 K)
= 50 degC / 1 K / (30 degC / 1 K)
= 1.066
Temperature scaling of the gas volume: maximum over minimum absolute temperature

[22] V0 = V0_ideal * f_rg * f_T
= 29.697 L × 1 × 1.066
= 31.656 L
Pre-charge gas volume V0 (accumulator size required)

[23] T_ratio = (T_c / 1 K) / (T_max / 1 K)
= 20 degC / 1 K / (50 degC / 1 K)
= 0.90716
Charging over maximum absolute temperature

[24] p0 = pa_0 * T_ratio
= 90 bar × 0.90716
= 81.645 bar
Absolute pre-charge pressure at the charging temperature

[25] p0_min = pa_0 / f_T
= 90 bar / 1.066
= 84.43 bar
Absolute pre-charge pressure at the minimum operating temperature
[26] p0_set = p0
= 81.645 bar
= 81.645 bar
Pre-charge to set at the charging temperature (absolute)
[27] u_r0 = pa_0 / (r_0 * pa_1)
= 90 bar / (0.9 × 100 bar)
= 1
Pre-charge against the p0/p1 rule
[28] u_rp = (pa_2 / p0_min) / r_p
= 210 bar / 84.43 bar / 4
= 0.62182
Pressure ratio p2/p0 (p0 at the minimum temperature) against the rule
[29] V_sel = smallest listed size not below V0
= V0 = 31.656 L, 4 sizes listed
= 32 L
Selected accumulator size
[30] u_V = V0 / V_sel
= 31.656 L / 32 L
= 0.98926
Required volume over the selected size
[31] bar_litres = PS * V_sel
= 230 bar × 32 L
= 7360 bar·L
Pressure-volume product PS·V (PS as entered, V the selected size)
[32] BL_pssr = 250 bar*L PSSR Schedule 1
= 250 bar·L
PSSR Schedule 1 exception threshold (indication)

5. Results

SYMBOL	VALUE	DESCRIPTION
V0	31.656 L	Accumulator gas volume required
V_sel	32 L	Selected size
u_V	0.98926	Selected size utilisation
p0	81.645 bar	Pre-charge, absolute, at the charging temperature
p0_set	81.645 bar	Pre-charge to set at the charging temperature
p0_min	84.43 bar	Pre-charge, absolute, at the minimum operating temperature
f_T	1.066	Gas volume scaling T_max/T_min
u_r0	1	Pre-charge rule utilisation (p0/p1)
u_rp	0.62182	Pressure ratio utilisation (p2/p0 at the minimum temperature)
bar_litres	7360 bar·L	Pressure-volume product (bar·litres)

SYMBOL	VALUE	DESCRIPTION
pssr	"7360 bar·L is at or above 250 bar·L: the exception does not apply; PSSR and PED / UK PER duties follow."	PSSR Schedule 1 250 bar·L exception (indication only)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pre-charge within the p0/p1 rule	u_r0 = 1	u_r0 ≤ 1	1	accumulator.rules record
PASS	Pressure ratio within the p2/p0 rule	u_rp = 0.62182	u_rp ≤ 1	0.622	accumulator.rules record

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: r_0, r_p (Pre-charge rules); sizes (Accumulator size list). Enter each as one so its source, edition and licence travel with it.
WARNING — realgas: p_2: the highest working pressure is 210 bar absolute, above 200 bar; nitrogen departs from the ideal-gas law there and the volume shown is the ideal-gas value. Enter a real-gas factor f_rg from a gas-property source.

8. Verification

Template fp.accumulator@1.2.0 Content hash 13736352fb1985ad Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0